CS 70 离散数学 和 概率论
笔记 10
春季 2026 Course Notes
To Infinity 和 Beyond: Countability 和 Computability
The next two notes tie together two topics that may at first sight appear to have nothing to do with each other: the nature of infinity (和 more particularly, the distinction between different levels of infinity), 和 thefundamentalnatureofcomputationandproof. Inthisfirstnotewewillfocusonnotionsofinfinity,和 thenturntoconnectionswithcomputationinthesecondnote.
1 Bijections

Twofinitesetshavethesamesizeifandonlyiftheirelementscanbepairedup,sothateachelementofone sethasauniquepartnerintheotherset,andviceversa. Weformalizethisthroughtheconceptofabijection.
考虑 a 函数 (或 mapping) f that maps elements of a 集合 A (called the 定义域 of f) to elements of setB(calledtherangeof f). 因为 f isafunction,itmustspecify,foreachelementx∈A(“input”),exactly one 元素 f(x)∈B (“output”). 回忆 we write this as f :A→B. We say that f is a bijection 如果 every 元素 a∈A has a unique image b= f(a)∈B, 和 every 元素 b∈B has a unique pre-image a∈A: f(a)=b.
f isaone-to-onefunction(oraninjection)如果 f mapsdistinctinputstodistinctoutputs. Moreprecisely, f is one-to-oneifthefollowingholds: (x∈A)(y∈A)(x =y = f(x) = f(y)).
f isonto(orsurjective)ifit“hits”everyelementintherange(i.e.,eachelementintherangehasatleastone pre-image). Moreprecisely,afunction f isontoifthefollowingholds: (y∈B)(x∈A)(f(x)=y).

Herearesomesimpleexamplestohelpvisualizeone-to-oneandontofunctions,andbijections:
One-to-one Onto Both(bijection)
1 a 1 a 1 a
b 2 2 b
2 b
c 3 3 c
3 d 4 c 4 d
A B A B A B
Notethat,accordingtotheabovedefinitions, f isabijectionifandonlyifitisbothone-to-oneandonto.
练习. 令 f :A→Bbeabijection. Showthat f hasaninverse f 1:B→Athatsatisfies f 1(f(a))=a foralla∈A,andthat f 1 isalsoabijection.

2 Cardinality

Howcanwedeterminewhethertwosetshavethesamecardinality(或“size”)? Theanswertothisquestion, reassuringly,liesinearlygradeschoolmemories: bydemonstratingapairingbetweenelementsofthetwo sets. More formally, we need to demonstrate a bijection f between the two sets. The bijection sets up a

one-to-onecorrespondence,orpairing,betweenelementsofthetwosets. We’veseenabovehowthisworks

forfinitesets.
Intherestofthislecture,wewillseewhatittellsusaboutinfinitesets.
Our first 问题 about 无限的 sets is the following: Are there more natural numbers N than 存在 positiveintegersZ+?
Itistemptingtoansweryes,sinceeverypositiveintegerisalsoanaturalnumber,但是 thenaturalnumbershaveoneextraelement0∈/Z+.
Uponmorecarefulobservation,though,weseethatwe

candefineabijectionbetweenthenaturalnumbersandthepositiveintegersasfollows:

N 0 1 2 3 4 5 ...
f
Z+ 1 2 3 4 5 6 ...

More precisely, this mapping is defined by f(n)=n+1 对所有 n∈N.
Why is this mapping a bijection?
Clearly, the 函数 f :N→Z+ is onto because every 正的 整数 is hit.
和 it is also one-to-one becausenotwo naturalnumbershavethe sameimage. (Theimageofnis f(n)=n+1, 所以 如果 f(n)= f(m) 那么 we must have n=m.) 因为 我们有 shown a bijection between N 和 Z+, this tells us that 存在 asmanynaturalnumbersastherearepositiveintegers!
(Very)informally,wehaveprovedthat“∞+1=∞.”
What about the 集合 of even natural numbers 2N={0,2,4,6,...}?
In the previous 例子 the difference wasjustoneelement.
Butinthisexample,thereseemtobetwiceasmanynaturalnumbersasthereareeven naturalnumbers.
Surely,thecardinalityofNmustbelargerthanthatof2NsinceNcontainsalloftheodd naturalnumbersaswell?
Thoughitmightseemtobeamoredifficulttask,letusattempttofindabijection betweenthetwosetsusingthefollowingmapping, f(n)=2nforalln∈N:

N 0 1 2 3 4 5 ...
f
2N 0 2 4 6 8 10 ...

The mapping in this 例子 is also a bijection. f is clearly one-to-one, 因为 distinct natural numbers get mappedtodistinctevennaturalnumbers(since2n=2mimpliesn=m). f isalsoonto,sinceeveryninthe rangeishit: itspre-imageis n.
Sincewehavefoundabijectionbetweenthesetwosets,thistellsusthatin
2
factNand2Nhavethesamecardinality!
What about the 集合 of all integers, Z? At first glance, it may seem obvious that the 集合 of integers is larger

thanthesetofnaturalnumbers,sinceitincludesinfinitelymanynegativenumbers.
然而,asitturnsout,

itispossibletofindabijectionbetweenthetwosets,meaningthatthetwosetshavethesamesize!
考虑
thefollowingmapping f:

... ...
0 1 2 3 4 124
0 1 1 2 2 ... 62 ...

Inotherwords,ourfunctionisdefinedasfollows:
x
ifxiseven

2
f(x)=
x+1
ifxisodd
2
We will 证明 that this 函数 f :N→Z is a bijection, by first showing that it is one-to-one 和 那么 showingthatitisonto.
证明 (one-to-one): 假设 f(x) = f(y). 那么 they both must have the same sign. 因此 either
f(x)= x 和 f(y)= y,或 f(x)= x+1 和 f(y)= y+1. Inthefirstcase,
2 2 2 2
x y
f(x)= f(y) = = = x=y.
2 2
Inthesecondcase,
x+1 y+1
f(x)= f(y) = = = x=y.
2 2
Soinbothcases f(x)= f(y) = x=y,所以 f isinjective.
证明(onto): Ify∈Zisnon-负的, 那么 f(2y)=y. 因此, yhasapre-image.
Ifyisnegative, 那么
f((2y+1))=y. 因此,yhasapre-image. Theseveryy∈Zhasapreimage,所以
f isonto.
因为 f isabijection,thistellsusthatNandZhavethesamesize.
Nowforanimportantdefinition.
Definition10.1.
WesaythatasetSiscountableifthereisabijectionbetweenSandNorsomesubsetofN.
因此 any 有限的 集合 S is countable (因为 存在 a bijection between S 和 the 子集 {0,1,2,...,m 1},
wherem=|S|isthesizeofS).
Andwehavealreadyseenthreeexamplesofcountableinfinitesets: Z+
和

2Nareobviouslycountablesincetheyarethemselvessubsetsof N;andZiscountablebecausewehavejust seenabijectionbetweenitandN.
What about the 集合 of all 有理数 numbers? 回忆 Q={x |x,y∈Z, y =0}.
Surely 存在 more
y
有理数 numbers than natural numbers? After all, 存在 infinitely many 有理数 numbers between any two natural numbers. Surprisingly, the two sets have the same cardinality! To see this, 令 us introduce a slightlydifferentwayofcomparingthecardinalityoftwosets.
Ifthereisaone-to-onefunction f
:A→B,thenthecardinalityofAislessthanorequaltothatofB. Nowto

showthatthecardinalityofAandBarethesamewecanshowthat|A|≤|B|和|B|≤|A|.
Thiscorresponds
toshowingthatthereisaone-to-onefunction f
:A→Bandaone-to-onefunctiong:B→A. Theexistence

ofthesetwoone-to-onefunctionsimpliesthatthereisabijectionh:A→B,thusshowingthatAandBhave thesamecardinality. Theproofofthisfact,
whichiscalledtheCantor-Bernsteintheorem, isactuallyquite hard,andwewillskipithere.
Back to comparing the natural numbers 和 the integers. First it is obvious that |N|≤|Q| because N Q.

所以 our goal now is to 证明 that also $|Q| \leq |N|$. To do this, we must exhibit an injection $f : Q \to N$. The followingpictureofaspiralconveystheideaofthisinjection:

3
2
1
3 2 1 1 2 3
1
2
3

Eachrationalnumber $\frac{a}{b}$ (writteninitslowestterms,sothatgcd(a,b)=1)isrepresentedbythepoint(a,b) in the 无限的 two-dimensional grid shown (其中 corresponds to $Z \times Z$, the 集合 of all pairs of integers).
笔记 that 非 all points on the grid are valid representations of rationals: e.g., all points on the x-axis have b=0 所以 none are valid (except 对于 (0,0), 其中 we take to represent the 有理数 number 0); 和 points suchas(2,8)和(1, 4)arenotvalideitherastherationalnumber $\frac{1}{4}$ isrepresentedby(1,4). But$Z \times Z$

certainlycontainsallrationalsunderthisrepresentation,soifwecomeupwithaninjectionfrom$Z \times Z$to$N$ thenthiswillalsobeaninjectionfrom$Q$to$N$(why?).

Theideaistomapeachpair(a,b)toitspositionalongthespiral,startingattheorigin.
(因此,e.g.,$(0,0) \to 0$,
$(1,0) \to 1$, $(1,1) \to 2$, $(0,1) \to 3$,andsoon.)
Itshouldbeclearthatthismappingmapseverypairofintegers

injectivelytoanaturalnumber,becauseeachpairoccupiesauniquepositionalongthespiral.
Thistellsusthat$|Q| \leq |N|$.
Sincealso$|N| \leq |Q|$,asweobservedearlier,bytheCantor-BernsteinTheorem
NandQhavethesamecardinality.
练习.
Showthattheset$N \times N$ofallorderedpairsofnaturalnumbersiscountable.
Our next 例子 concerns the 集合 of all binary strings (of any 有限的 length), denoted $\{0,1\}^*$. Despite the

factthatthissetcontainsstringsofunboundedlength,itturnsouttohavethesamecardinalityasN.
Tosee
this,wesetupadirectbijection $f : \{0,1\}^* \to N$asfollows.
Notethatitsufficestoenumeratetheelements
of$\{0,1\}^*$
insuchawaythateachstringappearsexactlyonceinthelist.
Wethengetourbijectionbysetting
f(n)tobethenthstringinthelist.
Howdoweenumeratethestringsin$\{0,1\}^*$?
Well,it'snaturaltolistthem

inincreasingorderoflength,andthen(say)inlexicographicorder(或,equivalently,numericallyincreasing

orderwhenviewedasbinarynumbers)withinthestringsofeachlength.
Thismeansthatthelistwouldlook
like
ε,0,1,00,01,10,11,000,001,010,011,100,101,110,111,0000,...
whereε denotestheemptystring(theonlystringoflength0).
Itshouldbeclearthatthislistcontainseach

binarystringonceandonlyonce,sowegetabijectionwithNasdesired.

Ourfinalcountableexampleisthesetofallpolynomialswithnaturalnumbercoefficients,whichwedenote N(x). To see that this 集合 is countable, we will make use of (a variant of) the previous 例子. 笔记 first

that,byessentiallythesameargumentasfor{0,1} ,wecanseethatthesetofallternarystrings{0,1,2} (thatis,stringsoverthealphabet{0,1,2})iscountable. ToseethatN(x)iscountable,itthereforesufficesto exhibitaninjection f :N(x)→{0,1,2} ,whichinturnwillgiveaninjectionfromN(x)toN. (Itisobvious that 存在 an injection from N to N(x), 因为 each natural number n is itself trivially a 多项式, namelytheconstantpolynomialnitself.)

Howdowedefine f? 令’sfirstconsideranexample,namelythepolynomialp(x)=5x5+2x4+7x3+4x+6. Wecanlistthecoefficientsof p(x)asfollows: (5,2,7,0,4,6). Wecanthenwritethesecoefficientsasbinary strings: (101,10,111,0,100,110). Now, we can construct a ternary string 其中 a “2” is inserted as a

separatorbetweeneachbinarycoefficient(ignoringcoefficientsthatare0). Thuswemap p(x)toaternary stringasillustratedbelow:

5x5+2x4+7x3+4x+6

1012102111221002110

Itiseasytocheckthatthisisaninjection,sincetheoriginalpolynomialcanbeuniquelyrecoveredfromthis ternary string by simply reading off the coefficients between each successive pair of 2’s. (Notice that this mapping f :N(x)→{0,1,2} isnotonto(andhencenotabijection)sincemanyternarystringswillnotbe

theimageofanypolynomials;thiswillbethecase,forexample,foranyternarystringsthatcontainbinary subsequenceswithleadingzeros.) HencewehaveaninjectionfromN(x)toN,soN(x)iscountable.

3 Cantor’s Diagonalization

我们有 established that N, Z, Q all have the same cardinality. What about R, the 集合 of 实数 numbers? Surelytheyarecountabletoo? Afterall,therationalnumbers,liketherealnumbers,aredense(i.e.,between any two 有理数 numbers a,b 存在 a 有理数 number, 即 a+b). In fact, between any two 实数

2

numbersthereisalwaysarationalnumber. Itisreallysurprising,那么,thattherearemorerealnumbersthan rationals. Thatis,thereisnobijectionbetweentherationals(orthenaturalnumbers)andthereals. Weshall

nowprovethis,usingabeautifulargumentduetoCantorthatisknownasdiagonalization. Infact,wewill showsomethingevenstronger: therealnumbersintheinterval[0,1]areuncountable!

练习.

Showhowtofindarationalnumberbetweenanytwo(distinct)realnumbers.

In preparation 对于 the 证明, 回忆 any 实数 number can be written out uniquely as an 无限的 decimal

with no trailing zeros. In particular, a 实数 number in the
interval [0,1] can be written as 0.d d d .... In
1 2 3

thisrepresentation,wewriteforexample1as0.999...1,and0.5as0.4999....
(Thusrationalnumberswill

alwaysberepresentedasrecurringdecimals,whileirrationaloneswillberepresentedasnon-recurringones.

Importantlyforus,alloftheseexpressionswillbeinfinitelylongandunique.)
Theorem10.1.
TherealintervalR[0,1](andhencealsothesetofrealnumbersR)isuncountable.
证明. 假设 towards a 矛盾 that 存在 a bijection f :N→R[0,1]. 那么, we
can enumerate
therealnumbersinaninfinitelist f(0),f(1),f(2),...asfollows:
f(0)= 0 . 5 2 1 4 9 3 5 6 ...
f(1)= 0 . 1 4 1 6 2 9 8 5 ...
f(2)= 0 . 9 4 7 8 2 7 1 2 ...
f(3)= 0 . 5 3 0 9 8 1 7 5 ...
. .
. .
. .

1Toseethis,writex=.999....Then10x=9.999...,so9x=9,andthusx=1.

Now 考虑 the 实数 number r whose 无限的 decimal expansion is 给定 by
the diagonal elements in the
表 above (circled in the diagram): 即, r = 0.5479.... We use r
to construct a new 实数 number s
obtained by modifying every digit of r, say by replacing
each digit d with d+2 (模 10); 因此 in our
exampleabove,s=0.7691....
Weclaimthatsdoesnotoccurinourinfinitelistofrealnumbers. 假设
对于 矛盾 that it did, 和 that it was the nth number in the list,
f(n). 但是 通过构造 s differs
from f(n) in the (n+1)th digit, 所以 these two numbers cannot
be equal! 所以 我们有 constructed a 实数
number s 即 非 in the 值域 of f. 但是 this contradicts the
assertion that f is a bijection. 因此 the 实数
numbersarenotcountable.
令 us 注 that the reason that we modified each digit by adding
2 (模 10)as opposed to adding 1 is
that the same 实数 number can have two decimal expansions; 对于
例子 0.999...=1.000.... 但是 如果 two
实数 numbers differ by more than 1 in any digit they cannot be
equal. 因此 we are completely safe in our
assertion. (An alternative way of avoiding this potential
pitfall is to replace each digit by some different
digitchosenfromtherange{1,2,...,8}.)
With Cantor’s diagonalization method, we proved that R is
uncountable. 因为 我们有 obtained dramati-
cally different results 对于 R 和 对于 Q (the first is
uncountable, the second is countable), we should really

gobackandconsiderwhatgoeswrongifweapplythediagonalizationmethodtoQ,inanattempttoshow

therationalsintheinterval[0,1]areuncountableWell,supposeforcontradictionthatourbijectivefunction
f :N→Q[0,1]producesthefollowingmapping:
f(0)= 0 . 1 4 0 0 0 ...

f(1)= 0 . 5 9 2 4 5 ...
f(2)= 0 . 2 1 4 2 1 ...
. .
. .
. .

Thistime,letusconsiderthenumberqobtainedbymodifyingeverydigitofthediagonal,saybyreplacing
eachdigitd withd+2 (模 10).
Thenintheaboveexampleq=0.316...,andwewanttotrytoshowthat
itdoesnotoccurinourinfinitelistofrationalnumbers.
然而,wedonotknowthatqisrational(infact,
it is extremely unlikely 对于 the decimal expansion of q to be
periodic). This is why the method fails when
appliedtotherationals.
Whendealingwiththereals,themodifieddiagonalnumberwasguaranteedtobea
realnumber.
4 The Cantor 集合
The Cantor 集合 is a remarkable 集合 construction involving the
实数 numbers in the interval [0,1]. The 集合 is
defined by repeatedly removing the middle thirds of line
segments infinitely many times, starting with the
original interval. 对于 例子, the first iteration would involve
the removal of the (open) interval (1,2),
3 3
leaving[0,1]∪[2,1].
Wethenproceedtoremovethemiddlethirdofeachofthesetworemainingintervals,
3 3
andsoon. Thefirstthreeiterationsareillustratedbelow:
0 1
1)
2)
3)
The Cantor 集合 contains all points that have 非 been removed:
C ={x :x notremoved}. How much of
the original unit interval is left after this process is
repeated infinitely? Well, we start with an interval of

length1,andafterthefirstiterationweremove 1
ofit,leavinguswith 2. Fortheseconditeration,wekeep
3 3
2×2 oftheoriginalinterval.
Aswerepeattheseiterationsinfinitelyoften,weareleftwith:
3 3
(cid:18) (cid:19)n
2 2 2 2 2 2 2
1 → → × → × × → • • • → lim =0
3 3 3 3 3 3 n→∞ 3

Accordingtothecalculations,wehaveremovedeverythingfromtheoriginalinterval!
Doesthismeanthat
the Cantor 集合 is empty? No, it doesn’t. What it means is
that the measure of the Cantor 集合 is zero; the
Cantor 集合 consists of isolated points 和 does 非 contain any
non-trivial intervals. In fact, 非 only is the
Cantorsetnon-empty,itisuncountable!2
Toseewhy,letusfirstmakeafewobservationsaboutternarystrings.
Internarynotation,allstringsconsist
of digits (called “trits”) from the 集合 {0,1,2}. All 实数
numbers in the interval [0,1] can be written in

ternary notation. (E.g., 1 can be written as 0.1, 或
equivalently as 0.0222..., 和 2 can be written as 0.2
3 3
或 as 0.1222....) 因此, in the first iteration, the middle
third removed contains all ternary numbers of the
form 0.1xxxxx. The ternary numbers left after the first
removal can all be expressed either in the form
0.0xxxxx... 或 0.2xxxxx...(我们有 to be a little careful here
with the endpoints of the intervals; 但是 we
can handle them by writing 1 as 0.02222... 和 2 as 0.2.) The
second iteration removes ternary numbers
3 3
of the form 0.01xxxxx 和 0.21xxxxx (i.e., any number with 1
in the second position). The third iteration
removes 1's in the third position, 和 所以 on. 因此, what remains
is all ternary numbers with only 0's
and2's. Thuswehaveshownthat

C={x∈[0,1]:xhasaternaryrepresentationconsistingonlyof0'sand2's}.
Finally,usingthischaracterization,wecansetupanontomap f
fromCto[0,1]. Sincewealreadyknowthat
[0,1]isuncountable,thisimpliesthatCisuncountablealso. Themap
f isdefinedasfollows: forx∈C, f(x)

isdefinedasthebinarydecimalobtainedbydividingeachdigitoftheternaryrepresentationofxby2.
因此,
对于 例子, 如果 x=0.0220 (in ternary), 那么 f(x) is the binary
decimal 0.0110. 但是 the 集合 of all binary
decimals 0.xxxxx... is in 1-1 correspondence with the 实数
interval [0,1], 和 the map f is onto because
everybinarydecimalistheimageofsometernarystringunder f
(obtainedbydoublingeverybinarydigit).3
ThiscompletestheproofthatCisuncountable.

5 Power Sets 和 Higher Orders of Infinity

LetSbeanyset. Thenthepowerset
ofS,denotedbyP(S),isthesetofallsubsetsofS. Moreformally,it
is defined as: P(S)={T :T S}. 对于 例子, 如果 S={1,2,3}, 那么
P(S)={{},{1},{2},{3},{1,2},
{1,3},{2,3},{1,2,3}}.
What is the cardinality of P(S)? 如果 |S|=k is 有限的, 那么
|P(S)|=2k. To see this, 令 us think of each

subsetofScorrespondingtoakbitstring,wherea1intheithpositionindicatesthattheithelementofSis
inthesubset,anda0indicatesthatitisnot.
Intheexampleabove,thesubset{1,3}correspondstothestring
101. Now the number of binary strings of length k is 2k, 因为
存在 two choices 对于 each bit position.
因此|P(S)|=2k. SoforfinitesetsS,thecardinalityofthepowersetofS
isexponentiallylargerthanthe

2It's actually easy to see that C contains at least
countably many points, 即 the endpoints of the intervals in
the
construction—i.e., numbers such as 1, 2, 1, 1 etc. It's less
obvious thatC also contains various other points, such as 1
和
3 3 9 27 4
3.(Why?)

3Notethat f isnotinjective;
forexample,theternarystrings0.20222...and0.22maptobinarystrings0.10111...and0.11
respectively,whichdenotethesamerealnumber. 因此 f
isnotabijection. 然而,thecurrentproofshowsthatthecardinality

ofCisatleastthatof[0,1],whileitisobviousthatthecardinalityofCisatmostthatof[0,1]sinceC [0,1].
HenceChasthe
samecardinalityas[0,1](andasR).

cardinality of S. What about 无限的 (countable) sets? We claim
that 存在 no bijection from S to P(S),
soP(S)isnotcountable.
Thusforexamplethesetofallsubsetsofnaturalnumbersisnotcountable,even
thoughthesetofnaturalnumbersitselfiscountable.
Theorem10.2. |P(N)|>|N|.
证明. Supposetowardsacontradictionthatthereisabijection f
:N→P(N). Recallthatwecanrepresent
a 子集 by a binary string, with one bit 对于 each 元素 of N. (所以,
因为 N is 无限的, the string will be
infinitely long. Contrast the case of {0,1} discussed
earlier, 其中 consists of all binary strings of 有限的
length.) 考虑 the following diagonalization picture in 其中 the
函数 f maps natural numbers x to
binarystringswhichcorrespondtosubsetsofN:
...
0 1 2 3 4 5
...
0 1 0 0 0 0 0
...
1 0 0 0 0 0 0
...
2 1 0 1 0 0 0
.
.
.
In this 例子, 我们有 assigned the following mapping: 0→{0}, 1→{},
2→{0,2}, ... (i.e., the nth
row describes the nth 子集 as follows: 如果 存在 a 1 in the kth
column, 那么 k is in this 子集, else it is
非.) Using a similar diagonalization argument to the earlier
one, flip each bit along the diagonal: 1→0,
0→1,andletbdenotetheresultingbinarystring.
First,wemustshowthatthenewelementisasubsetofN.

Clearlyitis,sincebisaninfinitebinarystringwhichcorrespondstoasubsetofN.
Nowsupposebwerethe
nthbinarystring.
Thiscannotbethecasethough,sincethenthbitofbdiffersfromthenthbitofthediagonal
(the bits are flipped). 所以 it’s 非 on our list, 但是 it should
be, 因为 we assumed that the list enumerated all
possiblesubsetsofN.
Thuswehaveacontradiction,implyingthatP(N)isuncountable.
因此 我们有 seen that the cardinality of P(N) (the power 集合 of
the natural numbers) is strictly larger
thanthecardinalityofNitself. ThecardinalityofNisdenoted
(pronounced“alephnull”),whilethatof
0
P(N)isdenoted2 0.
ItturnsoutthatinfactP(N)hasthesamecardinalityasR(therealnumbers),
和

indeed as the 实数 numbers in [0,1]. This cardinality is known
as c, the “cardinality of the continuum.” 所以
weknowthat2 0 =c>
0
.
Evenlargerinfinitecardinalities(或“ordersofinfinity”),denoted
1
,
2
,...,
can be defined using the machinery of 集合 theory; these obey
(to the uninitiated somewhat bizarre) rules of
arithmetic.
Severalfundamentalquestionsinmodernmathematicsconcerntheseobjects.
Forexample, the
famous“continuumhypothesis”assertsthatc=
(whichisequivalenttosayingthattherearenosetswith
1

cardinalitystrictlybetweenthatofthenaturalnumbersandthatoftherealnumbers).